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1)

Programiz C Online Compiler

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main.c

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Run

Output

Clear

```
1 #include <stdio.h>
2 int main() {
3     int a[3][3], b[3][3], sum[3][3], i, j, r=3, col=3;
4     printf("Enter elements of first matrix:\n");
5     for(i = 0; i < r; i++) {
6         for(j = 0; j < col; j++) {
7             scanf("%d", &a[i][j]);
8         }
9     }
10    printf("Enter elements of second matrix:\n");
11    for(i = 0; i < r; i++) {
12        for(j = 0; j < col; j++) {
13            scanf("%d", &b[i][j]);
14        }
15    }
16    for(i = 0; i < r; i++) {
17        for(j = 0; j < col; j++) {
18            sum[i][j] = a[i][j] + b[i][j];
19        }
20    }
21    printf("\nSum of the matrices:\n");
22    for(i = 0; i < r; i++) {
23        for(j = 0; j < col; j++) {
24            printf("%d ", sum[i][j]);
25        }
26        printf("\n");
27    }
28    return 0;
29 }
```

```
Enter elements of first matrix:
0 0 0 1 1 1 2 2 2
Enter elements of second matrix:
1 1 1 2 2 2 3 3 3

Sum of the matrices:
1 1 1
3 3 3
5 5 5

=== Code Execution Successful ===
```

2)

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Output

Clear

```
1 #include <stdio.h>
2
3 int main() {
4     int a[100], n, i;
5     printf("Enter the number of elements: ");
6     scanf("%d", &n);
7     printf("Enter the array elements:\n");
8     for(i = 0; i < n; i++) {
9         scanf("%d", &a[i]);
10    }
11    printf("Array elements in reverse order:\n");
12    for(i = n - 1; i >= 0; i--) {
13        printf("%d ", a[i]);
14    }
15
16    return 0;
17 }
```

```
Enter the number of elements: 3
Enter the array elements:
4 5 6
Array elements in reverse order:
6 5 4

=== Code Execution Successful ===
```

3)

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```
1 #include <stdio.h>
2 int main() {
3     int a[100], n, i, j, count = 0;
4     printf("Enter the number of elements: ");
5     scanf("%d", &n);
6     printf("Enter the array elements:\n");
7     for(i = 0; i < n; i++) {
8         scanf("%d", &a[i]);
9     }
10    for(i = 0; i < n; i++) {
11        for(j = i + 1; j < n; j++) {
12            if(a[i] == a[j]) {
13                count++;
14                break;
15            }
16        }
17    }
18    printf("Total number of duplicate elements = %d", count);
19    return 0;
20 }
```

Output

Clear

Enter the number of elements: 5
Enter the array elements:
1 1 2 2 6
Total number of duplicate elements = 2

=== Code Execution Successful ===

4)

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Run

Clear

```
1 #include <stdio.h>
2 int main() {
3     int a[100], n, i, j, count;
4
5     printf("Enter the number of elements: ");
6     scanf("%d", &n);
7
8     printf("Enter the array elements:\n");
9     for(i = 0; i < n; i++) {
10         scanf("%d", &a[i]);
11     }
12
13     printf("Unique elements in the array are:\n");
14     for(i = 0; i < n; i++) {
15         count = 0;
16         for(j = 0; j < n; j++) {
17             if(a[i] == a[j]) {
18                 count++;
19             }
20         }
21         if(count == 1) {
22             printf("%d ", a[i]);
23         }
24     }
25
26     return 0;
27 }
```

Enter the number of elements: 6

Enter the array elements:

1 1 2 3 4 6

Unique elements in the array are:

2 3 4 6

=== Code Execution Successful ===

5)

main.c



Share

Run

Output

Clear

```
1 #include <stdio.h>
2
3 int main() {
4     int a[100], n, i, element;
5     printf("Enter the number of elements: ");
6     scanf("%d", &n);
7
8     printf("Enter the sorted array elements:\n");
9     for(i = 0; i < n; i++) {
10         scanf("%d", &a[i]);
11     }
12
13     printf("Enter the element to insert: ");
14     scanf("%d", &element);
15
16     i = n - 1;
17     while(i >= 0 && a[i] > element) {
18         a[i + 1] = a[i];
19         i--;
20     }
21     a[i + 1] = element;
22     n++;
23     printf("Array after insertion:\n");
24     for(i = 0; i < n; i++) {
25         printf("%d ", a[i]);
26     }
27     return 0;
28 }
```

Enter the number of elements: 5
Enter the sorted array elements:
12 24 56 78 89
Enter the element to insert: 34
Array after insertion:
12 24 34 56 78 89

=== Code Execution Successful ===

6)

main.c



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Run

Output

Clear

```
1 #include <stdio.h>
2
3- int main() {
4     int a[100], n, i, element, pos;
5     printf("Enter the number of elements: ");
6     scanf("%d", &n);
7
8     printf("Enter the array elements:\n");
9-     for(i = 0; i < n; i++) {
10         scanf("%d", &a[i]);
11     }
12
13     printf("Enter the position to insert: ");
14     scanf("%d", &pos);
15
16     printf("Enter the element to insert: ");
17     scanf("%d", &element);
18-     for(i = n - 1; i >= pos - 1; i--) {
19         a[i + 1] = a[i];
20     }
21     a[pos - 1] = element;
22     n++;
23     printf("Array after insertion:\n");
24-     for(i = 0; i < n; i++) {
25         printf("%d ", a[i]);
26     }
27     return 0;
28 }
```

Enter the number of elements: 4
Enter the array elements:
12 23 11 23
Enter the position to insert: 2
Enter the element to insert: 15
Array after insertion:
12 15 23 11 23

=== Code Execution Successful ===

7)

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Output

Clear

```
1 #include <stdio.h>
2 int main() {
3     int a[3][3], b[3][3], c[3][3];
4     int i, j, k;
5     printf("Enter elements of first matrix:\n");
6     for(i = 0; i < 3; i++) {
7         for(j = 0; j < 3; j++) {
8             scanf("%d", &a[i][j]);
9         }
10    }
11    printf("Enter elements of second matrix:\n");
12    for(i = 0; i < 3; i++) {
13        for(j = 0; j < 3; j++) {
14            scanf("%d", &b[i][j]);
15        }
16    }
17    for(i = 0; i < 3; i++) {
18        for(j = 0; j < 3; j++) {
19            c[i][j] = 0;
20            for(k = 0; k < 3; k++) {
21                c[i][j] = c[i][j] + a[i][k] * b[k][j];
22            }
23        }
24    }
25    printf("Product of the matrices:\n");
26    for(i = 0; i < 3; i++) {
27        for(j = 0; j < 3; j++) {
28            printf("%d ", c[i][j]);
29        }
30        printf("\n");
31    }
32    return 0;
33 }
```

Enter elements of first matrix:

1 2 3
3 4 5
8 7 6

Enter elements of second matrix:

2 3 9
3 4 5
1 6 7

Product of the matrices:

11 29 40
23 55 82
43 88 149

--- Code Execution Successful ---

8)

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Run

Output

Clear

```
1 #include <stdio.h>
2
3 int main() {
4     int a[3][3], i, j;
5
6     printf("Enter elements of the matrix:\n");
7     for(i = 0; i < 3; i++) {
8         for(j = 0; j < 3; j++) {
9             scanf("%d", &a[i][j]);
10        }
11    }
12
13    printf("Transpose of the matrix:\n");
14    for(i = 0; i < 3; i++) {
15        for(j = 0; j < 3; j++) {
16            printf("%d ", a[j][i]);
17        }
18        printf("\n");
19    }
20
21    return 0;
22 }
```

```
Enter elements of first matrix:
1 2 3
3 4 5
8 7 6
Enter elements of second matrix:
2 3 9
3 4 5
1 6 7
Product of the matrices:
11 29 40
23 55 82
43 88 149

=== Code Execution Successful ===
```


9)

```
main.c  Run  Output  Clear
1  #include <stdio.h>
2
3- int main() {
4      int a[3][3], i, j, sum = 0;
5
6      printf("Enter elements of the matrix:\n");
7-      for(i = 0; i < 3; i++) {
8-          for(j = 0; j < 3; j++) {
9              scanf("%d", &a[i][j]);
10             }
11         }
12
13-      for(i = 0; i < 3; i++) {
14          sum = sum + a[i][2 - i];
15      }
16
17      printf("Right diagonal sum = %d", sum);
18
19      return 0;
20 }
```

```
Enter elements of the matrix:
4 5 6
2 3 4
8 7 9
Right diagonal sum = 17

=== Code Execution Successful ===
```

10)

Programiz

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Share

Run

```
1 #include <stdio.h>
2
3 int main() {
4     int a[3][3], i, j, flag = 1;
5     printf("Enter elements of the matrix:\n");
6     for(i = 0; i < 3; i++) {
7         for(j = 0; j < 3; j++) {
8             scanf("%d", &a[i][j]);
9         }
10    }
11
12    for(i = 0; i < 3; i++) {
13        for(j = 0; j < 3; j++) {
14            if(i == j && a[i][j] != 1) {
15                flag = 0;
16            }
17            else if(i != j && a[i][j] != 0) {
18                flag = 0;
19            }
20        }
21    }
22
23    if(flag == 1)
24        printf("The matrix is an identity matrix");
25    else
26        printf("The matrix is not an identity matrix");
27
28    return 0;
29 }
```

Output

Clear

```
Enter elements of the matrix:
1 0 0
0 1 0
0 0 1
The matrix is an identity matrix
=== Code Execution Successful ===
```

11)

Programiz

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Share

Run

Output

Clear

```
1 #include <stdio.h>
2
3 int main() {
4     char str[100];
5     int i, vowels = 0, consonants = 0, digits = 0, spaces = 0;
6     char ch;
7     printf("Enter a string: ");
8     fgets(str, sizeof(str), stdin);
9
10    for(i = 0; str[i] != '\0'; i++) {
11        ch = str[i];
12
13        if(ch == 'a' || ch == 'e' || ch == 'i' || ch == 'o' || ch == 'u' ||
14           ch == 'A' || ch == 'E' || ch == 'I' || ch == 'O' || ch == 'U') {
15            vowels++;
16        }
17        else if((ch >= 'a' && ch <= 'z') || (ch >= 'A' && ch <= 'Z')) {
18            consonants++;
19        }
20        else if(ch >= '0' && ch <= '9') {
21            digits++;
22        }
23        else if(ch == ' ') {
24            spaces++;
25        }
26    }
27    printf("Number of vowels = %d\n", vowels);
28    printf("Number of consonants = %d\n", consonants);
29    printf("Number of digits = %d\n", digits);
30    printf("Number of white spaces = %d\n", spaces);
31
32    return 0;
33 }
```

Enter a string: hello welcome 1239

Number of vowels = 5

Number of consonants = 7

Number of digits = 4

Number of white spaces = 2

--- Code Execution Successful ---

12)

main.c	Run	Output
<pre>1 #include <stdio.h> 2 3- int main() { 4 char str[100]; 5 int freq[256] = {0}; 6 int i; 7 printf("Enter a string: "); 8 fgets(str, sizeof(str), stdin); 9- for(i = 0; str[i] != '\0'; i++) { 10 freq[(unsigned char)str[i]]++; 11 } 12 printf("Frequency of characters:\n"); 13- for(i = 0; i < 256; i++) { 14- if(freq[i] > 0 && i != '\n') { 15 printf("%c = %d\n", i, freq[i]); 16 } 17 } 18 19 return 0; 20 }</pre>		<pre>Enter a string: internet Frequency of characters: e = 2 i = 1 n = 2 r = 1 t = 2 === Code Execution Successful ===</pre>

13)

 C Online Compiler

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main.c

Share

Run

Output

Clear

```
1 #include <stdio.h>
2
3 int main() {
4     char str[100];
5     int i, words = 0;
6     printf("Enter a string: ");
7     fgets(str, sizeof(str), stdin);
8     for(i = 0; str[i] != '\0'; i++) {
9         if((str[i] != ' ' && str[i] != '\n' && str[i] != '\t') &&
10            (i == 0 || str[i-1] == ' ' || str[i-1] == '\n' ||
11             str[i-1] == '\t')) {
12             words++;
13         }
14     }
15     printf("Number of words = %d", words);
16
17     return 0;
18 }
```

Enter a string: welcome to home
Number of words = 3

=== Code Execution Successful ===

14)

 Online Compiler

main.c

 Share

Run

Output

Clear

```
1 #include <stdio.h>
2 #include <string.h>
3 int main() {
4     char str[10][50], temp[50];
5     int n, i, j;
6     printf("Enter number of strings: ");
7     scanf("%d", &n);
8     printf("Enter the strings:\n");
9     for(i = 0; i < n; i++) {
10         scanf("%s", str[i]);
11     }
12     for(i = 0; i < n - 1; i++) {
13         for(j = i + 1; j < n; j++) {
14             if(strcmp(str[i], str[j]) > 0) {
15                 strcpy(temp, str[i]);
16                 strcpy(str[i], str[j]);
17                 strcpy(str[j], temp);
18             }
19         }
20     }
21     printf("Sorted strings are:\n");
22     for(i = 0; i < n; i++) {
23         printf("%s\n", str[i]);
24     }
25     return 0;
26 }
```

Enter number of strings: 3
Enter the strings:
Virat
Rahul
Steyn
Sorted strings are:
Rahul
Steyn
Virat

=== Code Execution Successful ===

Programiz PRO >

15)

 Online Compiler

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main.c

Share

Run

Output

Clear

```
1 #include <stdio.h>
2
3- int main() {
4     char str[100];
5     int i;
6     printf("Enter a sentence: ");
7     fgets(str, sizeof(str), stdin);
8-     for(i = 0; str[i] != '\0'; i++) {
9-         if(str[i] >= 'a' && str[i] <= 'z') {
10             str[i] = str[i] - 32;
11         }
12-         else if(str[i] >= 'A' && str[i] <= 'Z') {
13             str[i] = str[i] + 32;
14         }
15     }
16     printf("Toggled sentence: %s", str);
17
18     return 0;
19 }
```

Enter a sentence: HELLO hi
Toggled sentence: hello HI

=== Code Execution Successful ===